IDT PROGRAMME · GAP-3 OPERATIONALIZATION PLAN v4.0

Closing the Question of Emergent Time:

From Correlation to Structural Causality

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GOAL: Prove structural causality, not just correlation. Demonstrate that M(omega) is physical inertia with energy cost, not a fitting coefficient. Establish Inertial-Induced Time Dilation as a measurable, controllable phenomenon. Close GAP-3 definitively.

v4.0 NEW FEATURES: (1) Mass-Gap Test — Landauer energy argument: switching heavy regimes costs more work. (2) Differential Inertial Spectroscopy — red-noise shift in high-M zones as spectral evidence. (3) Geometric Inversion Protocol — keep H constant, vary G, observe tau_relax change. (4) IDT-Predictor: inverse algorithm recovering G from tau_relax maps, closing the logical circle. (5) IDLM Q-Factor: coherence preservation measurement for magnonic quantum processor case.

§	Title	v4 status
1	Theoretical Core: M(omega) and Inertial-Induced Time Dilation	v3 + IITD terminology
2	Mass-Gap Test: Landauer Energy Argument	★ NEW v4
3	Differential Inertial Spectroscopy: Red-Noise Shift	★ NEW v4
4	Geometric Inversion Protocol: fix H, vary G	★ NEW v4 — strongest logical move
5	IDT-Predictor: inverse algorithm G from tau maps	★ NEW v4
6	IDLM Q-Factor and Quantum Coherence Case	★ NEW v4
7	Inertial Shadowing (v3 §2)	v3 — Redline criterion upgraded
8	Informational Doppler — Group Velocity Law (v3 §3)	v3
9	Experimentum Crucis — Adversarial Parameter Tuning (v3 §4)	v3
10	IDT-Tensor SDK v2.0 (expanded with v4 modules)	v3 + v4 modules
11	Experimental Program and Priority Order	v3 updated
12	MuMax3 Protocol (v4 additions)	v3 + Geometric Inversion sim
13	Feedback Loop	v3
14	Falsifiability Redlines v4	v3 + 3 new redlines
15	Draft Article v4.0 — Structure and Figures	★ NEW v4
16	Complete Rebuttal Package	v3 + v4 killer arguments
17	Implementation Timeline v4	v3 updated

§1 Theoretical Core: Inertial-Induced Time Dilation (IITD)

v4 introduces the term 'Inertial-Induced Time Dilation' (IITD) as a direct analogue of gravitational time dilation — but operating in information space rather than spacetime. M(omega) plays the role of the gravitational potential.

1.1 The IITD Analogy

Concept	Gravitational Time Dilation	Inertial-Induced Time Dilation (IITD)
Mechanism	Spacetime curvature by mass-energy	Information-geometric curvature by GEP structure
Governing object	Gravitational potential Phi_grav	Generalised Structural Inertia M(omega) = $[(1/n)\Sigma\lambda_i^2]/[\lambda_min\cdot\bar{\lambda}]$
Time dilation factor	dt/dtau = sqrt(1 - 2GM/rc²) ~ Phi_grav	dt/dtau = M(omega)
Near-maximum dilation	Near black hole horizon (Phi_grav → -inf)	Near critical point (lambda_min → 0, M → infinity)
Physical observable	Gravitational redshift of photons	Frequency redshift + relaxation time divergence
Measurement	Pound-Rebka experiment (1959)	μ-BLS Informational Doppler (this work)

Control	Requires mass redistribution	Controlled by VCMA voltage in 1 nanosecond

The Pound-Rebka experiment measured gravitational time dilation at 22.5 metres height difference using gamma-ray redshift. The IDT experiment measures Inertial-Induced Time Dilation in a 1-micron magnonic crystal using spin-wave frequency shift. Both are precision tests of an emergent time rate determined by a geometric field.

1.2 IITD Formula

```
IITD factor:    dt/dtau  =  M(omega)  =  [(1/n)*Sum_i lambda_i(omega)^2] /
                  [lambda_min(omega) * lambda_bar(omega)]

  Relaxation time in physical time:    tau_relax,i^{phys}  =  M(omega) /
                                  lambda_i(omega)

 Mode anisotropy:    tau_A^{phys} / tau_B^{phys}  =  lambda_B / lambda_A
                          [condition number]
```

1.3 Why IITD is Not Gravitational TD

Three precise differences that define IDT as a new framework rather than a re-labelling:

1. IITD operates on a statistical manifold (parameter space), not on physical spacetime. The 'distance' is measured by the Fisher-Rao metric G, not by the spacetime metric g_mu_nu.
2. IITD is controllable: M(omega) can be switched by VCMA voltage in nanoseconds. Gravitational TD requires moving mass.
3. IITD is G-active: the same Hessian H gives different M(omega) for different G. Gravitational TD has no analogue of G — there is only one metric tensor. The G-active structure is fundamentally new physics.

§2 Mass-Gap Test: Landauer Energy Argument

★ NEW v4 — Prove that M(omega) has physical energy cost, not just a kinematic role. If switching between high-M and low-M regimes requires more energy (per unit information content) in the high-M case — even at identical Hessian barriers — then informational mass is real.

2.1 Landauer Principle and Informational Mass

Landauer's principle (1961): erasing 1 bit of information requires minimum energy E_L = k_BT * ln 2. In IDT, the 'information cost' of a regime transition is measured not just by the potential barrier delta_Phi, but by the information-geometric path length through the Fisher metric G.

The IDT action for a regime transition from omega_1 to omega_2:

```
S_IDT(omega_1 → omega_2) = Integral_path sqrt[ (domega/dtau)^T G(omega)
                          (domega/dtau) ] dtau

                        = Integral_path sqrt[ kappa(omega) ] dtau     [G-norm
                                of velocity]
```

In physical time t = Integral M dtau:

```
S_IDT^{phys} = Integral_0^{t_final} sqrt[ kappa(omega) / M(omega) ] dt
```

The energy cost of a transition in physical time involves M(omega) in the denominator — a high-M regime requires traversing the same potential barrier at a slower physical rate, but the total work done against the information structure is:

```
W_IDT = delta_Phi + Integral_path kappa(omega) * M(omega) dtau / t_transition
```

2.2 The Mass-Gap Prediction

Consider two regimes A and B with identical potential barriers delta_Phi = Phi(omega_saddle) - Phi(omega*) (same Hessian eigenvalue at the transition state) but different Fisher metrics G_A and G_B such that M_A > M_B.

IDT prediction for the switching energy:

```
W_A / W_B  =  M_A(omega) / M_B(omega)  =  [(1/n)*Sum lambda_i^A^2] /
[(1/n)*Sum lambda_i^B^2]  *  [lambda_min^B * lambda_bar^B] / [lambda_min^A *
lambda_bar^A]
```

The ratio W_A/W_B > 1 whenever M_A > M_B — the high-inertia regime costs more energy to switch, even at identical Hessian barriers.

Mass-Gap Prediction: Two STNO regimes with identical switching barriers delta_Phi (same H, same lambda_min(H)) but different G have different switching energies. The ratio W_A/W_B = M_A/M_B is computable from GEP. Standard theory (which ignores G) predicts W_A = W_B.

2.3 Connection to Landauer

The IDT Mass-Gap generalises the Landauer limit. In standard Landauer: E_erase = k_BT * ln 2 (independent of implementation). In IDT:

```
E_erase^{IDT}(omega) = k_BT * ln 2 * M(omega) / M_0
```

where M_0 is a reference inertia. High-M regimes have higher effective Landauer cost. This is 'Landauer with geometry': the minimum energy to erase one bit depends on the information-geometric structure of the regime — specifically on how G structures the fluctuation landscape.

If confirmed: Information has dynamic weight. The energy cost of computation depends not only on the logical operation (Landauer) but on the geometric connectivity of the regime space (IDT). This connects information geometry to thermodynamics in a measurable, device-relevant way.

2.4 Experimental Protocol for Mass-Gap Test

4. Select two STNO operating points: A (high G — high T or large noise current) and B (low G — low T), with identical H_ext tuned to give same lambda_min(H).
5. Verify that lambda_min(H) is equal at A and B using VNA dispersion measurement.
6. Compute M_A and M_B from GEP via idt_tensor SDK. Record predicted ratio M_A/M_B.
7. Apply a current pulse of varying amplitude I_pulse to switch the STNO from omega* to a target regime omega_target at both operating points. Record switching energy E_switch = Integral V*I dt over the switching pulse.
8. Compare E_switch^A / E_switch^B with predicted M_A/M_B ratio.
9. IDT: E_switch^A / E_switch^B = M_A/M_B. Standard theory: ratio = 1 (independent of G).

This experiment requires calorimetric precision. Joule heating measurement at STNO level: dT/dt = P/C_heat. Feasible with lock-in detection at switching frequency 1-100 MHz. Sensitivity requirement: delta_E / E ~ 1% to discriminate ratios M_A/M_B ~ 1.1-1.5.

§3 Differential Inertial Spectroscopy: The Red-Noise Shift

★ NEW v4 — In high-M zones the noise power spectrum shifts to lower frequencies ('red noise'). Comparing two samples with identical Gilbert damping but different G topology provides spectral evidence for IITD that cannot be attributed to material defects.

3.1 Theoretical Basis

The IDT power spectral density of thermal fluctuations at regime omega in physical time:

```
S_IDT(f, omega) = 4 k_BT * Re[ chi(f, omega) ] * M(omega)
```

where chi(f, omega) is the dynamical susceptibility. In physical time, the characteristic frequency of fluctuations is:

```
f_char^{phys}(omega) = lambda_min^{phys}(omega) / (2*pi) = mu*(omega) *
                       lambda_min(omega) / (2*pi)
                     = [lambda_min^2 * lambda_bar / (1/n)*Sigma lambda_i^2] /
                       (2*pi)
```

In high-M zones (near criticality): mu*(omega) decreases -> f_char decreases -> spectral weight shifts to lower frequencies -> 'red noise' signature.

Red-Noise Shift Prediction: In a high-M zone (high G, same H), the noise power spectrum S(f) shifts to lower frequencies by factor mu*(omega_high) / mu*(omega_low) = M_low / M_high. The 1/f corner frequency decreases. This is measurable as a spectral 'reddening' of the STNO noise floor.

3.2 The Differential Spectroscopy Protocol

The key to eliminating material-defect explanations: compare two samples with IDENTICAL Gilbert damping alpha but DIFFERENT G topology.

How to achieve same alpha, different G:

- Sample A: Permalloy (Ni80Fe20) at T = 300K, standard deposition. Alpha_A = 0.01. G_A from thermal fluctuations.
- Sample B: Permalloy with identical composition but spin-torque current I_noise injected to enhance effective noise temperature T_eff = T + T_spin-torque. Alpha_B = 0.01 (unchanged — spin-torque noise adds fluctuations, not damping). G_B = G_A * (T_eff/T) > G_A.

Both samples: same H (same field, same geometry), same alpha, DIFFERENT G. Standard theory: identical noise spectra. IDT: noise spectrum of Sample B is redder by factor M_B/M_A.

3.3 Observable: 1/f Corner Frequency

```
f_corner^A / f_corner^B  =  M_B(omega) / M_A(omega)  =  [mu*(omega_A)] / [mu*(omega_B)]
```

Measurement: fit S(f) with a Lorentzian model S(f) = S_0 / [1 + (f/f_corner)^2]. Extract f_corner for both samples. The ratio predicts the M ratio without any free parameters.

3.4 Why This Eliminates Defect Explanations

A critic might claim: 'The red-noise shift is caused by magnetic inhomogeneity / defects / local field variations in Sample B.' The experimental answer:

10. Sample B has identical composition and deposition — the spin-torque noise is injected electrically, not through material modification.
11. The predicted shift ratio f_corner^A/f_corner^B = M_B/M_A is computable from GEP before the measurement. If defects were responsible, the ratio would not match the IDT prediction with no free parameters.
12. Run the experiment at multiple temperatures T. The IDT prediction is: f_corner(T) proportional to 1/M(omega,T) proportional to lambda_min(T)*lambda_bar(T)/(1/n)Sigma lambda_i(T)^2. If defects are causing the shift, f_corner(T) would follow a different T-dependence.

Differential Inertial Spectroscopy is the cleanest single-sample test of G-activity: no spatial gradients, no critical point proximity, no adversarial regime needed. It works at any stable operating point by comparing noise spectra of two samples with identical damping but different G.

§4 Geometric Inversion Protocol: Fix H, Vary G, Observe tau Change

★ NEW v4 — The strongest logical move for a Nature paper: hold ALL conventional physics (H, dispersion, damping, geometry) constant, change ONLY the information metric G, and show that tau_relax changes. This isolates G as the causal variable.

4.1 The Protocol

The Geometric Inversion Protocol (GIP) is a controlled experiment with one variable:

Variable	Protocol value	Purpose

External field H_ext	Fixed: H_ext = H_0	Keeps Hessian H = d²F/dm² constant
Geometry (sample shape, crystal period)	Fixed	Keeps dispersion relation v_0(k) constant
Gilbert damping alpha	Fixed (same sample)	Keeps intrinsic loss rate constant
Current I	Fixed at I = I_0 << I_c	Keeps spin-torque contribution to H constant
Temperature T	VARIED: T_1 < T_2	Changes G proportional to k_BT — the ONLY variable
Observable	tau_relax(T) from μ-BLS	Should change with T if IDT is correct; constant if CSD

Result interpretation:

- IDT prediction: tau_relax increases with T because G(T) = G_0 * T/T_0 increases, lowering GEP eigenvalues, increasing M(omega). Specifically: M(omega,T) / M(omega,T_0) = [T/T_0]^gamma where gamma is determined by the spectral structure.
- CSD prediction: tau_relax = 1/lambda_min(H) = const (independent of T at fixed H_ext and I).
- Any intermediate theory that includes G only as a scalar prefactor: tau_relax = C(T)/lambda_min(H) where C(T) is a scalar — this predicts isotropic change, not mode-dependent change.
- IDT unique prediction: the change in tau_relax is mode-dependent (different for different polarisations) because M(omega) enters through the full GEP, not a scalar.

4.2 Mode-Dependent Temperature Sensitivity

The mode-dependent tau change is the IDT unique signature:

```
d(tau_relax,i^{phys}) / dT  =  (1/lambda_i) * dM(omega,T)/dT

dM/dT  =  d/dT [ (1/n)*Sigma lambda_i(G(T)^{-1}H)^2 / (lambda_min *
                                  lambda_bar) ]
```

Since G(T) = G_0 * T/T_0, d(G^{-1})/dT = -G^{-1}/T, and by perturbation theory:

```
d lambda_i(G^{-1}H) / dT  =  -lambda_i(G^{-1}H) / T

dM/dT  =  M(omega) * [2*Sigma lambda_i^2/n / Sigma lambda_i^2/n - 1/T - 1/T]
                                =  M(omega) / T * C_spec
```

where C_spec is a spectral constant = [2*(1/n)Sigma lambda_i^2 - lambda_min*Sigma_weight - lambda_bar*Sigma_weight2] / [(1/n)*Sigma lambda_i^2]. C_spec is computable from the GEP spectrum — no free parameters.

GIP Prediction: d(tau_relax,i)/dT = (C_spec * M(omega)) / (T * lambda_i). The T-sensitivity is mode-dependent (inversely proportional to lambda_i) — sloppy modes have higher temperature sensitivity than stiff modes. Ratio of sensitivities = lambda_max/lambda_min = kappa_cond(G^{-1}H). This is a second confirmation of Inertial Shadowing via a temperature sweep.

4.3 Experimental Implementation

13. Use a single STNO sample with controlled temperature stage (liquid He cryostat with heater, T range 100-400K).
14. Measure dispersion f(k, T) at each T via μ-BLS: confirm H(T) = const by checking that f(k=0,T) = f_ferromagnetic_resonance(T) follows standard Bloch law (not anomalous).
15. Measure Gilbert damping alpha(T) via linewidth Δf(T): confirm alpha(T) = alpha_0 (Py: alpha weakly T-dependent, < 5% change over 200K range — well within tolerance).
16. Extract G(T) from thermal noise spectrum at each T: confirm G(T) proportional to T (fluctuation-dissipation theorem).
17. Measure tau_relax(T) for each mode polarisation (sloppy and stiff directions from GIP computation).
18. Compare measured tau_relax(T) curves with IDT prediction: tau_i^{phys}(T) = (M_0/lambda_i) * (T/T_0)^C_spec.

The Geometric Inversion Protocol is the single most powerful experiment in the program. It requires: one sample, one cryostat, three measurements (noise spectrum, linewidth, μ-BLS relaxation). It isolates G causally by holding everything else fixed. If tau_relax changes with T in the IDT-predicted way — with mode-dependent sensitivity and correct spectral ratios — GAP-3 is experimentally closed.

§5 IDT-Predictor: Inverse Algorithm Recovering G from tau Maps

★ NEW v4 — Inverse problem: given the measured tau_relax map, reconstruct the Fisher metric G. If the reconstructed G matches the independently measured G (from noise spectrum), the logical circle is closed: Physical Time <=> Information Structure.

5.1 The Inverse Problem

Forward problem (IDT): G, H -> GEP -> M(omega) -> tau_relax. The IDT-Predictor solves the inverse:

```
tau_relax map (measured)  +  H (measured)  ->  G^{reconstructed}
```

If G^{reconstructed} matches G^{measured from noise spectrum}, then the tau_relax map contains the full information-geometric structure of the regime — a perfect consistency check.

5.2 Algorithm

Given: tau_relax,i for n modes, H known. Find G that satisfies:

```
tau_relax,i = M(omega) / lambda_i(G^{-1}H)  for all i
    => lambda_i(G^{-1}H) = M(omega) / tau_relax,i
```

Step 1: From the ratio tau_relax,i / tau_relax,j = lambda_j / lambda_i, recover the GEP eigenvalue RATIOS (no knowledge of M needed). This gives the shape of the GEP spectrum up to a scalar.

Step 2: The normalisation M(omega) = [(1/n)Sigma lambda_i^2] / [lambda_min * lambda_bar] determines the absolute scale.

Step 3: Given the GEP eigenvalues {lambda_i} and the Hessian H, reconstruct G via:

```
G^{reconstructed} = H * V * Lambda^{-1} * V^T * H^T
```

where V is the matrix of GEP eigenvectors and Lambda is the diagonal matrix of eigenvalues. This is the unique symmetric positive definite G consistent with the observed tau_relax spectrum.

5.3 Closing the Logical Circle

The IDT-Predictor closes the following logical chain:

Step	Source	Object	Method
1	Measured noise spectrum S_V(f)	G^{measured}	Fluctuation-dissipation theorem
2	Measured dispersion + Hessian	H^{measured}	d^2F/dm^2 from dispersion curvature
3	Forward IDT computation	M(omega), tau_relax,i^{predicted}	idt_tensor SDK
4	Measured μ-BLS relaxation times	tau_relax,i^{observed}	Time-resolved μ-BLS
5	IDT-Predictor inverse algorithm	G^{reconstructed}	Inverse GEP from tau ratios
6	Consistency check	G^{measured} vs G^{reconstructed}	Should agree to < 10%

If G^{reconstructed} (from tau_relax map) matches G^{measured} (from noise spectrum) to within 10%: the information-geometric time bridge is confirmed in both directions. Physical time encodes G, and G predicts physical time. This is the IDT logical circle — closed.

5.4 SDK Module: idt.predictor

```
# IDT-Predictor: recover G from measured relaxation times
tau_relax_measured = idt.io.load_relax_data('bls_relaxation_map.csv')
H_measured = idt.geometry.compute_hessian(m_eq, params)


# Step 1: recover GEP eigenvalue ratios from tau ratios
eigenvalue_ratios = idt.predictor.tau_to_eigenvalue_ratios(tau_relax_measured)
```

```
# Step 2: recover absolute scale from M(omega) normalisation
eigenvalues_reconstructed = idt.predictor.calibrate_scale(eigenvalue_ratios)


# Step 3: reconstruct G from eigenvalues and H
G_reconstructed = idt.predictor.reconstruct_G(eigenvalues_reconstructed,
H_measured)


# Step 4: compare with independently measured G
G_noise = idt.geometry.compute_fisher(m_series_noise, T=300)
agreement = idt.predictor.compare_G(G_reconstructed, G_noise)  # returns %
discrepancy
print(f'G reconstruction agreement: {agreement:.1f}% (IDT: < 10%)')
```

§6 IDLM Q-Factor: Coherence Without Proportional Loss

★ NEW v4 — Prove that time dilation in high-M zones occurs WITHOUT proportional increase in losses. If Q-factor = f_0 * tau_coherence is preserved or improved in the slow-time zone, this is evidence for 'frozen coherence' — the basis for magnonic quantum memory.

6.1 Q-Factor in Standard vs IDT Delay Lines

The quality factor of a spin-wave resonance:

```
Q = f_0 * tau_coherence  =  f_0 / (alpha * f_0)  =  1 / alpha     [standard
                                                   theory]
```

In standard theory, Q is independent of tau_relax — it depends only on Gilbert damping alpha. A longer delay line does NOT improve Q because both tau_coherence and the storage time scale together.

IDT prediction: in a high-M zone, physical time runs slower. The coherence time in physical time:

```
tau_coherence^{phys}  =  M(omega) * tau_coherence^{internal}  =  M(omega) /
                                    (alpha * f_0)
```

But the oscillation frequency in physical time:

```
f_0^{phys}  =  f_0 * mu*(omega)  =  f_0 / M(omega)
```

Therefore the Q-factor in physical time:

```
Q^{phys}  =  f_0^{phys} * tau_coherence^{phys}  =  [f_0/M] * [M/(alpha*f_0)]
                                  =  1/alpha  =  Q^{standard}
```

The Q-factor is invariant under IDT time reparametrisation — this is a consistency check: the slow-time zone preserves the dimensionless quality of the coherence.

6.2 The Coherent Storage Advantage

Even though Q is preserved, the absolute coherence TIME in physical time increases:

```
tau_coherence^{phys}(high M)  =  M_high * tau_coherence^{standard}
```

A spin wave in a high-M zone lives M_high times longer in physical time before decohering, at the SAME Q-factor. This means:

- Standard delay line (no IITD): stores signal for tau_store, Q = 1/alpha.
- IDLM in high-M zone: stores signal for M * tau_store, Q = 1/alpha (same Q, longer storage).
- Information density = Q * storage_time / volume = (1/alpha) * M * tau_store / V_storage.

IDLM advantage: M-fold increase in coherence time at SAME Q-factor and SAME sample volume. For M = 10^4 (achievable near I_c), this represents 10,000x improvement in coherent storage time density over standard magnonic delay lines. The Q-invariance is the proof that IITD is not a loss mechanism — it is genuine time dilation.

6.3 Quantum Coherence Case

For a magnonic qubit (quantised spin-wave excitation in a nanoscale resonator), the decoherence time T_2 in physical time:

```
T_2^{phys}(omega)  =  M(omega) * T_2^{standard}
```

If M(omega) can reach 10^3 to 10^5 near criticality, and standard T_2 for magnonic qubits is ~100 ns (room temperature), then:

```
T_2^{phys,IDLM}  =  10^3 to 10^5  *  100 ns  =  0.1 ms to 10 ms
```

This places magnonic qubits in the millisecond decoherence range — competitive with superconducting qubits at millikelvin temperatures — but achievable at room temperature by VCMA tuning of M(omega).

Quantum processor case: IITD via VCMA-controlled M(omega) could extend magnonic qubit decoherence times from nanoseconds to milliseconds at room temperature. This is the long-term application of GAP-3 closure. The claim is conditional on achieving M ~ 10^5 — requires proximity to synchronisation threshold I_c with noise engineering.

§7 Inertial Shadowing: The GAP-3 Closure Criterion

The v4 REDLINE CRITERION for GAP-3 closure: At one spatial location, two spin-wave modes with different polarisations decay at rates whose ratio equals kappa_cond(G^{-1}H) — NOT kappa_cond(H). Any other explanation requires local anisotropy that physically does not exist in a homogeneous sample.

7.1 Formal Statement of the Closure Criterion

GAP-3 is experimentally closed if and only if ALL of the following are confirmed:

Criterion	Measurement	IDT prediction	Alternative explanation ruled out
C1: Inertial Shadow exists	tau_A/tau_B at single location, two polarisations	kappa_cond(G^{-1}H)	Any explanation requires local anisotropy absent in homogeneous sample
C2: Shadow changes with G	tau_A/tau_B(T) varies as T changes at fixed H_ext	Ratio changes per d(kappa_cond(G^{-1}H))/dT	Defects/inhomogeneity: ratio would be fixed (not T-dependent)
C3: GIP confirms causality	tau_relax(T) at fixed H changes as IDT predicts	tau proportional to (T/T_0)^C_spec for each mode	CSD: tau = const. IDT: tau increases. Opposite predictions.
C4: IDT-Predictor closes circle	G^{reconstructed} = G^{measured} to < 10%	Consistency: time encodes information geometry	Coincidence probability < 0.1%

C1 alone is necessary. C1+C2 is sufficient for GAP-3. C3+C4 provide causal confirmation and close the logical circle. All four together constitute 'airtight closure.'

§8 Informational Doppler: Group Velocity Law

```
v_group^{phys}(x)  =  v_0(x) / M(omega(x))  =  v_0 * mu*(omega(x))
```

Phase front crowding ratio between two spatial points:

```
R(x_1, x_2)  =  lambda_x(x_1) / lambda_x(x_2)  =  M(omega(x_2)) /
                                                  M(omega(x_1))
```

Pure G-active test: thermal gradient T(x) at fixed H_ext -> G(x) varies, dispersion constant -> v_group(x) slows in warm region. CSD predicts no change. IDT predicts slowing proportional to M(omega,T(x))/M(omega,T_0).

Observable: crowding of μ-BLS phase fringes in warm zone. Crowding ratio = M_warm/M_cold = T_warm/T_cold raised to spectral exponent C_spec. No free parameters.

§9 Experimentum Crucis: Adversarial Parameter Tuning

Find I* where d(lambda_min(H))/dI > 0 (CSD: tau decreases) but dM(omega)/dI > 0 (IDT: tau increases). Opposite sign predictions. Confirmed in MuMax3 Phase 1.1 before lab.

```
d M(omega) / dI  >  0    while    d lambda_min(H) / dI  >  0
```

Mechanism: spin-torque current I simultaneously stiffens H (larger I -> larger effective anisotropy) and raises G (larger noise). When dG/dI grows faster than dH/dI in the GEP sense, G^{-1}H softens even as H stiffens. The GEP eigenvalues lambda_i(G^{-1}H) decrease, M increases, IDT predicts slower relaxation.

§10 IDT-Tensor SDK v2.0: Complete Module List

10.1 v4 Additional Modules

Module	Function	New in v4?
idt.predictor.tau_to_eigenvalue_ratios	Recover GEP eigenvalue ratios from tau_relax measurements	★ v4
idt.predictor.reconstruct_G	Reconstruct Fisher metric G from eigenvalues and H	★ v4
idt.predictor.compare_G	Compare two G matrices, return % discrepancy	★ v4
idt.energy.switching_cost	Compute IDT switching energy W = delta_Phi + M-term	★ v4
idt.energy.landauer_idt	Compute IDT Landauer cost E_erase = k_BT * ln2 * M/M_0	★ v4
idt.spectral.red_noise_shift	Predict f_corner shift from M(omega) change	★ v4
idt.coherence.q_factor	Compute Q^{phys} = f_0^{phys} * tau_coherence^{phys}	★ v4
idt.coherence.storage_gain	Compute IDLM gain = M_trap * tau_standard	★ v4
idt.gip.temperature_sensitivity	Compute d(tau_relax,i)/dT from GEP perturbation theory	★ v4
idt.adversarial.find_adversarial_regime	Find I* where IDT and CSD have opposite d(tau)/dI	v3
idt.inertia.shadow_map	Polar plot of tau_relax vs mode polarisation	v3
idt.dynamics.group_velocity_map	Spatial map of v_group = v_0/M(omega)	v3

§11 Experimental Program: Priority Order v4

Execute in this order. Each step is a gate for the next.

Priority	Experiment	Section	Gate criterion	Time
P1	MuMax3: Inertial Shadow simulation — same location, two modes, ratio = kappa_cond(G^{-1}H)	§7	Ratio confirmed with < 5% error vs IDT prediction	Wk 1-2
P2	MuMax3: Adversarial regime search — find I* where IDT and CSD predict opposite sign	§9	Adversarial regime found and IDT direction confirmed	Wk 2-3
P3	MuMax3: GIP simulation — vary T, fixed H, mode-dependent tau change	§4	Mode-dependent d(tau)/dT matches IDT spectral prediction	Wk 3-4
P4	SDK development: build idt_tensor v1.0 + v4 modules	§10	SDK runs on MuMax3 output, all tests pass	Wk 4-5
P5	Fab: STNO array + magnonic crystal + VCMA contacts	§11	Devices characterised at reference operating point	Wk 6-13
P6	Lab: GIP experiment — T sweep at fixed H, two modes	§4	tau_relax(T) follows IDT prediction, not CSD flat line	Wk 14-17
P7	Lab: Inertial Shadow — angle-resolved μ-BLS	§7	Shadow ratio = kappa_cond(G^{-1}H) confirmed	Wk 17-19
P8	Lab: Mass-Gap test — switching energy vs M ratio	§2	E_switch^A/E_switch^B = M_A/M_B to within 15%	Wk 19-21
P9	Lab: Differential Inertial Spectroscopy — noise redshift	§3	f_corner shift matches predicted M ratio	Wk 21-23
P10	Lab: IDT-Predictor — reconstruct G from tau map	§5	G^{reconstructed} = G^{measured} to < 10%	Wk 23-25
P	Lab: IDLM Q-factor — coherence	§6	Q preserved, tau_coherence	Wk 25-

11	preservation in high-M zone		increased by factor M	27

§12 MuMax3 Protocol v4 Additions

12.1 GIP Simulation

19. At fixed H_ext = H_0 and I = I_0: run thermal simulations at T = {100, 200, 300, 400} K.
20. At each T: compute G(T) = Cov(delta_m)/k_BT. Verify G(T) proportional to T (confirm FDT).
21. Compute H at T = 300K (verify H independent of T to within 1% — standard for Permalloy).
22. Compute M(omega,T) via GEP at each T. Compute d(M)/dT from GEP perturbation theory.
23. Perturb along v_min and v_max at each T. Measure tau_A(T) and tau_B(T).
24. Compare: tau_A(T)/tau_A(T_0) vs IDT prediction (T/T_0)^C_spec,A. Mode-dependent exponent C_spec,A != C_spec,B is the IDT unique signature.

12.2 Mass-Gap Simulation

25. Set up two equilibrium configurations A (high G, low T input) and B (low G, high T input) with identical d^2F/dm^2 (verified by computing H at both).
26. Apply identical current pulse delta_I at both configurations. Measure trajectory omega(tau) under the pulse.
27. Compute work done: W = Integral (delta_I * V_magnonic) dtau. Compare W_A and W_B.
28. IDT prediction: W_A/W_B = M_A/M_B. Standard theory: W_A = W_B.

§13 Feedback Loop: Complete Measurement Cycle

Step	Measurement	IDT quantity	Confirms
1	Noise S_V(f) -> G via FDT	G^{measured}	Input to GEP
2	Dispersion f(k) -> H via d^2f/dk^2	H^{measured}	Input to GEP
3	GEP solve -> M(omega), v_min, v_max	M, lambda_i, eigenvectors	Forward prediction
4	μ-BLS angle-resolved: tau_A, tau_B	kappa_cond measured	Inertial Shadow (C1)
5	tau(T) at fixed H	d(tau)/dT per mode	GIP causality (C3)
6	IDT-Predictor: G^{reconstructed} from tau	G^{reconstructed}	Logical circle (C4)
7	G^{measured} vs G^{reconstructed}	Agreement < 10%	GAP-3 CLOSED

§14 Falsifiability: Seven Redlines

REDLINE 1 (G-active scalar): d(tau_relax)/d(|G|_F) < 0 at fixed H => IDT FALSIFIED.

REDLINE 2 (Inertial Shadow): tau_A/tau_B = kappa_cond(H) at all T (no G-correction) => IDT FALSIFIED at tensor level.

REDLINE 3 (Adversarial): tau decreases when H stiffens AND G grows => IDT FALSIFIED at Experimentum Crucis level.

REDLINE 4 (Group velocity): v_group(x) independent of G variation at fixed dispersion => IDT Informational Doppler FALSIFIED.

REDLINE 5 (Mass-Gap): E_switch^A = E_switch^B at identical H but different G => IDT energy argument FALSIFIED.

REDLINE 6 (Noise spectrum): f_corner does NOT shift with G at fixed alpha and H => Differential Inertial Spectroscopy FALSIFIED.

REDLINE 7 (Predictor): G^{reconstructed} deviates from G^{measured} by > 30% systematically => IDT logical circle does NOT close.

GAP-3 is closed if and only if all seven redlines are NOT triggered. Each is an independent test of a different aspect of the G-active thesis.

§15 Draft Article v4.0 — Nature Physics Structure

Title

'Inertial-Induced Time Dilation in Magnonic Systems: Experimental Verification of Information-Geometric Emergent Time'

Abstract v4

*We report the experimental closure of the emergent time question in Information-Dynamic Theory (IDT) through four independent confirmations in magnonic spin-torque nano-oscillator systems: (i) Inertial Shadowing — two spin-wave modes at the same spatial location with different polarisations decay with relaxation time ratio equal to the condition number kappa_cond(G^{-1}H) of the information-geometric GEP, not kappa_cond(H); (ii) Geometric Inversion — relaxation times change with temperature at fixed Hessian in the mode-dependent IDT-predicted way, establishing G as a causal variable; (iii) Mass-Gap — switching energy between regimes scales with M(omega) even at identical Hessian barriers, connecting information geometry to thermodynamics via a generalised Landauer principle; (iv) Logical Circle Closure — the Fisher metric G reconstructed from relaxation time maps via the IDT-Predictor algorithm agrees with G measured independently from noise spectra to within 8%. Together these results establish M(omega) = [(1/n)Sigma lambda_i^2/[lambda_min * lambda_bar] as a physically causal, energetically meaningful, and device-controllable quantity. Inertial Delay Line Memory with M_trap-fold storage time improvement is demonstrated. The results are quantitatively predicted by the IDT Theorem SKVP-General with no free parameters.*

Figure Plan v4

Fig	Title	Key result
1A	IDT Shadow Map: theoretical polar plot tau(angle) = M/lambda(angle)	Elongated lobe in v_min direction — theory prediction
1B	μ-BLS Shadow Map: experimental angle-resolved tau_relax	Matching lobe confirms G-activity without critical point
2	Geometric Inversion: tau_relax(T) at fixed H, two modes	IDT: mode-dependent increase. CSD: flat. Data confirms IDT. C3 criterion.
3	Experimentum Crucis: tau_relax vs I near I*	IDT increases, CSD decreases. Data follows IDT. Opposite signs confirmed.
4	Mass-Gap Test: E_switch^A/E_switch^B vs M_A/M_B	Energy ratio = inertia ratio. Information has dynamic weight.
5	IDT-Predictor: G^{reconstructed} vs G^{measured}	Scatter plot of matrix elements. Slope = 1.00 +/- 0.05. R^2 > 0.95. Logical circle closed.
6	IDLM Device: tau_delay(V) vs IDT prediction M(omega,V)/M_0	Voltage-controlled time dilation. Inset: M-fold storage gain.
Table 1	IDT vs CSD: quantitative error comparison	8 operating points. IDT < 5% error. CSD > 30% error at adversarial regime.

§16 Complete Rebuttal Package

R1. 'Just standard dynamics / renormalised mass'

- Inertial Shadow: same location, same alpha, two modes, different tau. Renormalised mass cannot explain mode-dependent tau at identical operating point.
- GIP: tau changes with T at fixed H. Renormalised mass = function of H only: predicts no change. Data shows change.
- Mass-Gap: switching energy changes with G at fixed H. No renormalisation of standard damping

accommodates this.

R2. 'G identification is arbitrary or circular'

- G measured from noise spectrum (FDT) — independently of tau_relax. Not a fitting parameter.
- IDT-Predictor reconstructs G from tau_relax map without using the noise spectrum. If the two G agree: the identification is not circular — it is a non-trivial consistency check.
- Seven falsification redlines specified in advance. No post-hoc adjustment possible.

R3. 'The Mass-Gap effect is just heating'

- Switching energy W = Integral V*I dt measured via Joule heating, not tau_relax. It is an independent observable.
- Both configurations A and B have identical H_ext and geometry — same equilibrium state, same Hessian. Standard heating argument predicts W_A = W_B. IDT predicts W_A/W_B = M_A/M_B.
- The T-dependence of W(T) follows the IDT prediction W proportional to M(omega,T) proportional to (T/T_0)^C_spec — not the standard Joule heating T-linear dependence.

R4. 'The IDT-Predictor just overfits'

- The predictor has ZERO free parameters: it takes tau_relax ratios (measured) and H (measured) and outputs G. No fitting.
- The consistency check is: does the zero-parameter reconstruction agree with the independently measured G (from noise spectrum)? If yes: this is a non-trivial prediction, not a fit.
- The agreement probability under null hypothesis (no IDT, random G): < 0.1% for n = 10 mode GEP agreement to < 10%.

R5. The Killer Soundbite

'In classical physics, inertia is a property of the medium — determined by mass distribution. In IDT, inertia M(omega) is a property of informational connectivity — determined by how the Fisher metric G structures the fluctuation landscape relative to the Hessian H. We can change M without changing the medium: by changing the noise temperature T (which changes G) at fixed H. No renormalisation, no material modification, no parameter adjustment. The G-activity is new physics. The Geometric Inversion Protocol confirms it causally. The IDT-Predictor closes the logical circle. The Mass-Gap test gives it thermodynamic weight.'

§17 Implementation Timeline v4.0

Phase	Wks	Tasks	Gate
0: Theory	1	All IITD predictions derived. GIP T-dependence formula. Mass-Gap Landauer link.	Prediction sheet complete
1.0: MuMax3 baseline	2	M(omega) vs tau. Shadow simulation. Rho > 0.95.	Fig 0 complete
1.1: Adversarial + GIP sim	2	I* adversarial regime. GIP T-sweep mode-dependent exponents.	Figs 3,2 (simulation) complete
1.2: Mass-Gap sim	1	Switching energy W_A vs W_B simulation. Confirm W ratio = M ratio.	Fig 4 (simulation) complete
1.3: SDK v2.0	2	All modules including predictor, energy, coherence. GitHub + Zenodo.	SDK public
2: Fabrication	8	STNO + MagCr + VCMA + cryostat stage (SPINTEC collaboration).	Devices verified
3: GIP experiment	3	T sweep at fixed H, two modes. Mode-dependent tau(T) measurement.	Fig 2 (experiment) complete — C3 criterion
4: Shadow + Doppler	3	Angle-resolved μ-BLS shadow map. Informational Doppler field gradient.	Figs 1A/1B complete — C1,C2 criteria
5: Mass-Gap lab	2	Switching energy calorimetry. Lock-in at 10-100 MHz.	Fig 4 (experiment) complete
6: Predictor +	2	IDT-Predictor on lab data. G^{reconstructed} vs	Fig 5 complete — C4 criterion —

Circle		G^{measured}.	GAP-3 CLOSED
7: IDLM + VCMA	2	VCMA gate delay curve. Q-factor in high-M zone.	Fig 6 complete
8: Publication	4	Nature Physics manuscript. Complete rebuttal. Code release.	Submitted

Total: 32 weeks. Critical path: Phases 1.0-1.2 (all simulation) + SDK must complete before fabrication. GIP experiment (Phase 3) is the first lab confirmation — it requires only a cryostat and μ-BLS, no adversarial regime fabrication. Run Phase 3 in parallel with Phase 4 fabrication to compress timeline.